

Api 570 Piping Inspection

Page 1: Piping Classes and Inspection Intervals

API 570 covers inspection, rating, repair, and alteration of in-service metallic and fiberglass-reinforced plastic (FRP) piping systems. Piping systems are classified into three classes based on the hazard level of the process fluid: Class 1 (highest hazard: flammable gases, toxic chemicals, anhydrous acids), Class 2 (moderate hazard: hydrocarbons, low-pressure steam, sour water), and Class 3 (lowest hazard: non-flammable liquids, utility water, air).

Class 1 piping requires thickness measurement inspection every 5 years and visual inspection every 5 years. Class 2 requires thickness testing every 10 years, and Class 3 every 10 years for thickness with 15-year visual inspection. Thickness measurement is conducted using ultrasonic testing (UT) or profile radiography (RT).

Page 2: Corrosion Rates and Piping Integrity

Like API 510, corrosion rates are calculated as short-term or long-term. The thinnest point on a pipe spool dictates the remaining life of that entire circuit. Weld joints, elbows, and tees are highly susceptible to localized corrosion (such as erosion-corrosion or flow-accelerated corrosion) and must contain multiple TMLs.

Piping repairs must conform to API 570 requirements. Temporary repairs (such as split sleeves or leak clamps) must be replaced with permanent welded repairs during the next scheduled shutdown.